Fiscal Year:	FY 2026	**Task Last Updated:**	FY 05/12/2026
PI Name:	Sicienski, Shane Ph.D.		
Project Title:	Generation of Synthetic Data to Train Medical Machine Learning (ML) Algorithms		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	shane.sicienski@snke.com	**Fax:**	FY
PI Organization Type:	INDUSTRY	**Phone:**	716-380-0225
Organization Name:	Snke, LLC		
PI Address 1:	5 Westbrook Corporate Center		
PI Address 2:	Suite 1000		
PI Web Page:			
City:	Westchester	**State:**	IL
Zip Code:	60154-5766	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	TRISH--Catalyst
Start Date:	03/01/2025	**End Date:**	02/28/2026
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	TRISH
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNX16AO69A-CAT0007		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	The primary goal of this project is to leverage synthetic ultrasound imagery to train a machine learning (ML) algorithm capable of identifying jugular venous thrombosis autonomously, thereby addressing a critical need for improved diagnostic capabilities during space missions. This project seeks to address risks identified by the NASA Human Research Program (HRP) roadmap by developing an innovative solution for the early detection of jugular vein thrombosis (JVT), thereby enhancing the autonomy and safety of future space exploration endeavors. Specific Aims 1. Generate a dataset of synthetic ultrasound images of the internal jugular vein (IJV), based on a volumetric anatomical model of the neck. This dataset will vary parameters such as thrombus location and other relevant factors, drawing upon clinical knowledge and the unique physiological considerations of spaceflight to create a comprehensive and varied		

	dataset. 2. Train a suitable machine learning algorithm using the synthetic dataset to accurately detect IJV thrombosis. This aim involves developing and validating a ML model that can effectively identify JVT from ultrasound images, potentially incorporating real-time analysis capabilities to support autonomous diagnostic processes during space missions.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work informs best practices for using synthetic data to address out-of-distribution challenges in medical machine learning, including scenarios where models must operate in novel conditions such as spaceflight. More broadly, the work shows how targeted synthetic augmentation can be used to mitigate data scarcity and bias.
Task Progress:	This work demonstrates that targeted synthetic data can be used to mitigate training dataset bias in medical machine learning contexts. This provides insight into designing machine learning algorithms for novel environmental conditions like microgravity, which risk out-of-distribution problems if they are trained on non-representative data. We generated synthetic ultrasound data with specific anatomical variation for the IJV and used it to augment an ablated clinical training dataset designed to simulate a paucity of data. We observed modest but consistent improvements in segmentation algorithms trained on the augmented data. Results suggest that the relevance of the synthetic variation matters more than quantity, and that even low-fidelity simulations can be used to improve performance for settings where data is scarce, including but not limited to those relevant to space health.
Bibliography Type:	Description: (Last Updated:)